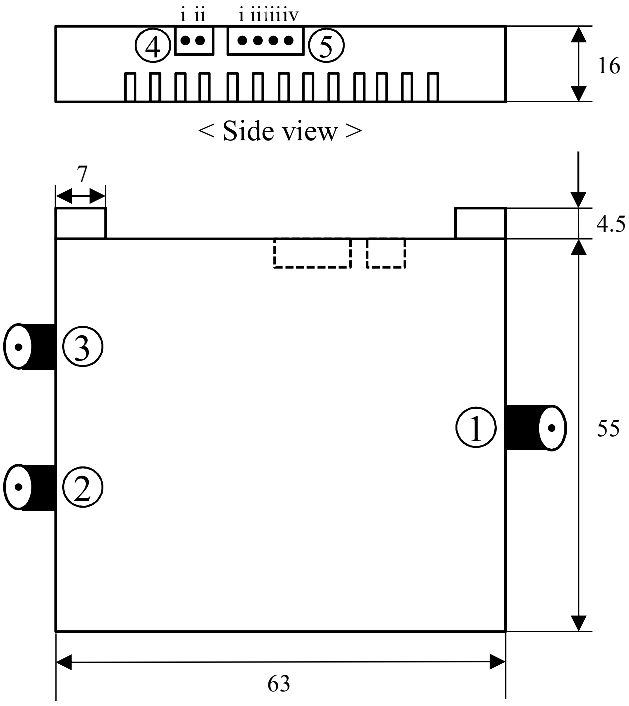
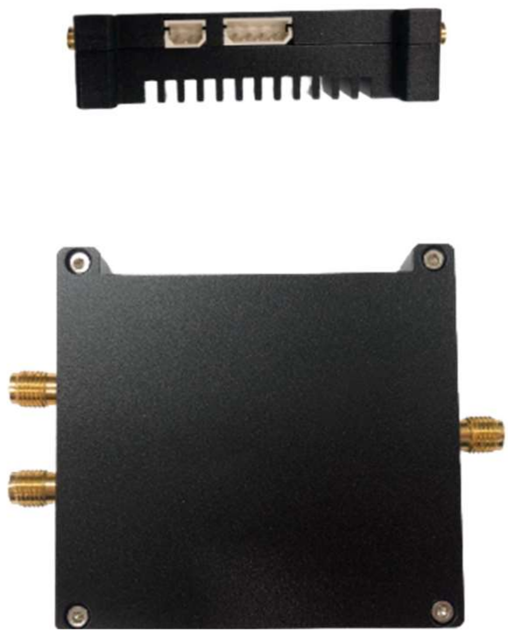
Mini-flat w/ heatsink & EVB Manual

A. Mechanical Dimensions





Tolerances : ±1 mm

< Picture >

B. Properties

No.	Name	Explanation
①	Input gate pulse	Gate pulse signal, ex) 10 MHz / 6.6 V / 2 ns
②	Output signal	Mini-flat output signal detect 1
③	Output signal for DA-SPAD or Balanced dual SPAD	Mini-flat output signal detect 2 (When using DA-SPAD, output is reference signal)
④	DC power	Setting the DC voltage according to mini-flat spec. Caution) Don't exceed 2V/sec when increasing voltage
		④-i: Bias(GND) / ④-ii: Bias(+)
⑤	TEC power & thermistor	TEC : 3MC06-024-05.TC222 (RMT) Thermistor : VH10-6E103F (MITSUBISHI, 10kΩ)
		⑤-i: thermistor1 / ⑤-ii: TEC(-) / ⑤-iii: TEC(+) / ⑤-iv: thermistor2

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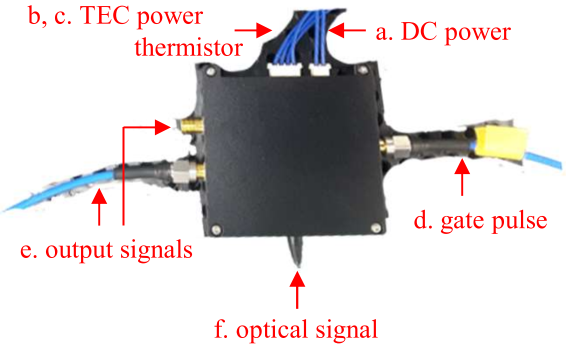
C. How to use

1. Initial setup – check the connection

a. DC power / b. TEC power / c. TEC thermistor

d. Gate pulse / e. Output signals / f. Optical signal / g. Recommended) cooling fan



2. Operation flow

a. Turn on the TEC power and set the temperature (ex. -40 °C)

b. Turn on the DC power

b-1. Setting the DC voltage according to mini-flat module spec.

b-2. Example) Breakdown voltage (V_B) = 70 V @ 25 °C

→ Temperature dependence of V_B = 0.11 V/ °C

→ 70 - 7.15 = 62.85 V → Start measuring at 56V (Consider gate pulse)

b-3. Caution) Don't exceed 2V/sec when increasing voltage

c. Input the gate pulse & optical signal

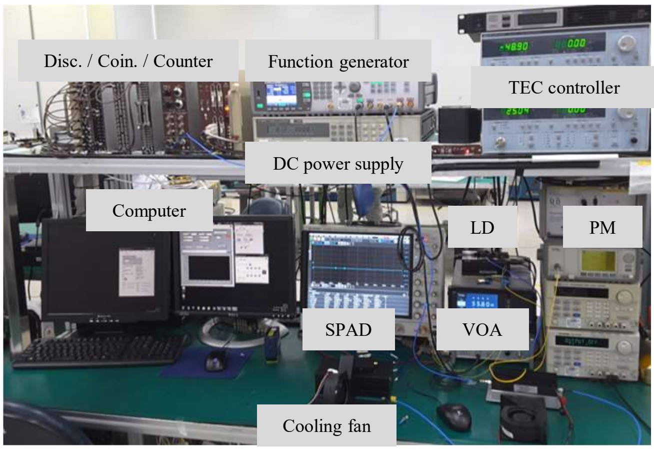
c-1. Example) Gate pulse: 10 MHz / 6.6 V / 2 ns

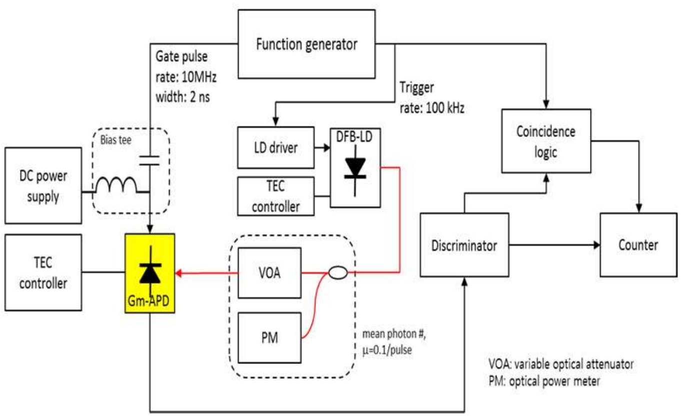
Optical signal: 100 KHz / 0.1 photon/pulse

3. Measurement

a. Check the output signal & calculate PDE, DCR, APP etc.

a-1. Example) Setup

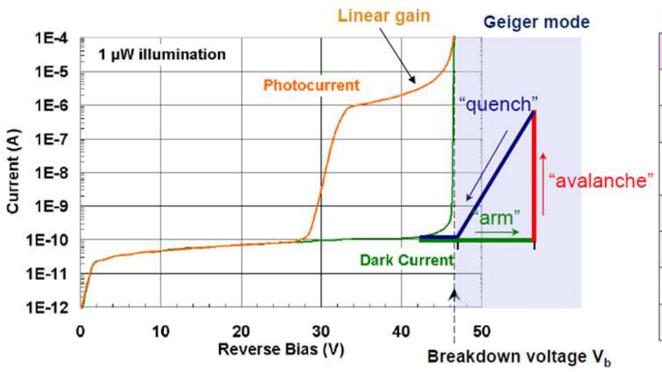




Appendix

A. Geiger-mode operation

- Self-sustaining avalanche is generated by applying a bias voltage higher than the breakdown voltage to the APD



< Characteristics of APD's linear mode & Geiger-mode >

Optical Characteristics (Tc=25°C)

Parameter	Symbol	Test Condition	Min	Typ.	Max	Unit
Breakdown Voltage	V_{BR}	ID=100uA	60	70	80	V
Total dark current	I_D	V_R = 0.98V_{BR}		1		nA
Capacitance	C_{PD}	f = 1MHz, V_{PD} = 0.9VB		0.1		pF
Quantum efficiency	η	M=1, 1550nm		70		%
Optical Wavelength Range	λ	-	1100		1600	nm
Responsivity	R	λ = 1550nm, M=1	0.7	0.8		A/W
Temperature coefficient of V_{BR}	Γ	$\Delta V_{BR}/\Delta T$		0.11		V/°C

< Optical Characteristics of Wooriro Mini-flat >

B. Geiger-mode APD performance parameters

- Photon detection efficiency (PDE)

$$P(n) = \frac{\mu^{\eta} e^{-\mu}}{n!} \rightarrow \eta = \frac{1}{\mu} ln\left(\frac{1 - R_{dark}/f_g}{1 - R^C/f_p}\right)$$

- Dark counts probability / gate (DCP)

$$P_{dark} = \frac{R_{dark}}{f_g}$$

- Afterpulse probability (APP)

$$P_{ap} = \frac{R_{de} - R^C - (1 - \frac{f_p}{f_g})R_{dark}}{R^C}$$

f_g : gate pulse rate

f_p : optical pulse rate

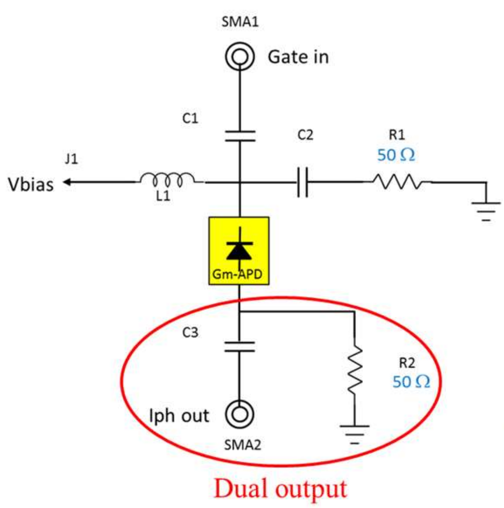
μ : mean photon number / pulse

R_{dark} : dark counts rate

R_{de}: total signal counts rate

R^C: coincidence rate

C. EVB Schematic diagram & TEC



TEC type	ΔTmax K	Qmax W	Imax A	Umax V	ACR Ohm
3MC06-024-05/Z2/v4.TC222	300K vacuum				
	113	0,6	2,4	1,95	0,69
	323K N2				
	115	0,66	2,4	2,2	0,77

< TEC datasheet >

10kΩ	Thermistor spec		
T (˚C)	Rmin (Ω)	Rnom (Ω)	Rmax (Ω)
-40	2.683E+05	2.812E+05	2.947E+05
-35	1.986E+05	2.075E+05	2.166E+05
-30	1.487E+05	1.548E+05	1.611E+05
-25	1.123E+05	1.165E+05	1.209E+05
-20	8.582E+04	8.876E+04	9.178E+04
-15	6.638E+04	6.844E+04	7.056E+04
-10	5.152E+04	5.297E+04	5.445E+04
-5	4.017E+04	4.119E+04	4.222E+04
0	3.140E+04	3.211E+04	3.282E+04
5	2.461E+04	2.509E+04	2.559E+04
10	1.943E+04	1.976E+04	2.010E+04
15	1.544E+04	1.567E+04	1.590E+04
20	1.233E+04	1.248E+04	1.264E+04
25	9.900E+03	1.000E+04	1.010E+04
30	7.959E+03	8.057E+03	8.156E+03
35	6.436E+03	6.529E+03	6.622E+03
40	5.233E+03	5.319E+03	5.407E+03
45	4.278E+03	4.357E+03	4.438E+03
50	3.516E+03	3.588E+03	3.661E+03

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